

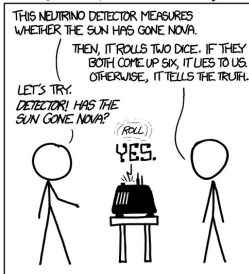
Econometrics I

Lecture 3a: Inference and Standard Errors

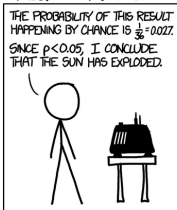
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DID THE SUN JUST EXPLODE? (IT'S NIGHT, SO WE'RE NOT SURE)



FREQUENTIST STATISTICIAN:



BAYESIAN STATISTICIAN:



Before the formulas: what are we uncertain about?

- ▶ Every inference question starts with an **estimand**: the number you would compute with infinite data. An elasticity. A treatment effect. The gender gap in log wages. (Goff's Appendix C calls this "defining the parameter of interest," and puts it *before* estimation.)
- ▶ A standard error quantifies exactly one kind of uncertainty: **sampling uncertainty** — you have n observations instead of all of them.
- ▶ It says *nothing* about the other ways to be wrong: omitted variables (last week), measurement error, the wrong functional form, the wrong estimand.
- ▶ Corollary you already simulated: the short regression in the Silly Example has a **tiny standard error around the wrong number**. Precision is not validity.

Today: how to get the sampling uncertainty right. Keep asking the other questions anyway.

Recap: Asymptotics for OLS and the Linear Model

$$y_i = \beta_0 + \beta x_i + u_i$$

Recall the three basic OLS assumptions

1. $\mathbb{E}(u_i | X_i) = 0$
2. $(X_i, Y_i), i = 1, \dots, n$, are i.i.d.
3. Large outliers are rare $\mathbb{E}[Y^4] < \infty$ and $\mathbb{E}[X^4] < \infty$.

Unbiasedness and Consistency

- Unbiasedness means on average we don't over or under estimate $\hat{\beta}$

$$\mathbb{E}[\hat{\beta}] - \beta_0 = 0$$

- Consistency tells us that we approach the true β_0 as $n \rightarrow \infty$.

$$\hat{\beta} \xrightarrow{p} \beta_0$$

- Example: $X_{(1)}$ is unbiased but not consistent for the mean.
- Example $\frac{n}{n-5} \overline{X}$ is consistent but biased for the mean.

- ▶ **Outliers** refer to observations that are “far away” from the rest of the data. They can be due to errors in the data. There is no standard formal definition.
- ▶ What to do? Greene: *“It is difficult to draw firm general conclusions... It remains likely that in very small samples, some caution and close scrutiny of the data are called for.”* I’d say that’s true even in large samples, but there isn’t a generally accepted way of quantifying what counts as appropriate “caution and close scrutiny.”

Removing Outliers?

- ▶ Removing extreme outliers (in x) from datasets is often considered good practice. But we should be mindful about why as dropping observations creates the potential for manipulation.
- ▶ Sometimes extreme outliers are just errors, in which case they should almost certainly be dropped.
- ▶ Even if they aren't errors, they may reflect a different mode in the data generating process. They may require a different or more general model to account for them properly. Consider the justification of a linear model based on Taylor's theorem (local linear approximation). With such a justification for your modeling strategy, it would not make sense to include an outlier in x .
- ▶ It's important to be transparent about how dropped outliers affect results.

- One way to find influential observations is to calculate the **leverage** of each observation i . We begin with the hat matrix:

$$P = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

and consider the diagonal elements, which are labeled h_{ii}

$$h_{ii} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i$$

- This tells us how influential an observation is in our estimate of $\mathbf{b}_{OLS}$. Particularly important for $\{0, 1\}$ dummy variables with uneven groups.

Leave One Out Regression

- ▶ This is sometimes called the **Jackknife**
- ▶ Sometimes it is helpful to know what would happen if we omitted a single observation i
- ▶ Turns out we don't need to run N regressions

$$\begin{aligned}\mathbf{b}_{-i} &= (\mathbf{X}'_{-i}\mathbf{X}_{-i})^{-1}\mathbf{X}'_{-i}\mathbf{y}_{-i} \\ &= \mathbf{b}_{OLS} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i\tilde{e}_i \quad \text{where } \tilde{e}_i = (1 - h_{ii})^{-1}e_i\end{aligned}$$

- ▶ $\tilde{e}_i$ has the interpretation of the LOO prediction error.
- ▶ high leverage observations move $\mathbf{b}_{OLS}$ a lot.

Bias Variance Decomposition

We can decompose any estimator into two components

$$\underbrace{\mathbb{E}[(y - \hat{f}(x))^2]}_{MSE} = \underbrace{\left(\mathbb{E}[\hat{f}(x) - f(x)]\right)^2}_{Bias^2} + \underbrace{\mathbb{E}\left[\left(\hat{f}(x) - \mathbb{E}[\hat{f}(x)]\right)^2\right]}_{Variance}$$

- What minimizes MSE?

$$f(x_i) = \mathbb{E}[Y_i \mid X_i]$$

- In general we face a tradeoff between bias and variance.
- In OLS we minimize the variance among unbiased estimators assuming that the true $f(x_i) = X_i\beta$ is linear. (But is it?)

- A useful identity for linear algebra:

$$\text{Var}(a \cdot \mathbf{Z}) = a^2 \cdot \mathbb{V}(\mathbf{Z})$$

$$\text{Var}(A \cdot \mathbf{Z}) = A \cdot \mathbb{V}(\mathbf{Z}) \cdot A'$$

- Since $\mathbf{b}_{OLS} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$,

$$\mathbb{V}(\mathbf{b}_{OLS} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \mathbb{V}(\mathbf{y} \mid \mathbf{X}) \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1}$$

- Recalling that $\mathbb{V}(\mathbf{y} \mid \mathbf{X}) = \mathbb{V}(\boldsymbol{\varepsilon} \mid \mathbf{X})$, (because $\mathbb{V}(\mathbf{X} \mid \mathbf{X}) = 0$)

$$\mathbb{V}(\mathbf{b}_{OLS} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \mathbb{V}(\boldsymbol{\varepsilon} \mid \mathbf{X}) \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1}$$

Start with the variance of the residuals to form a **diagonal** matrix D :

$$\text{Var}(\varepsilon|\mathbf{X}) = \mathbb{E}(\varepsilon\varepsilon' | \mathbf{X}) = \mathbf{D}$$

$$\mathbf{D} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2) = \begin{pmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n^2 \end{pmatrix}$$

- ▶ $\mathbf{D}$ is diagonal because $\mathbb{E}[\varepsilon_i \varepsilon_j | X] = \mathbb{E}[\varepsilon_i | x_i] \cdot \mathbb{E}[\varepsilon_j | x_j] = 0$ (independence)
- ▶ The elements of D_i are given by $\mathbb{E}[\varepsilon_i^2 | X] = \mathbb{E}[\varepsilon_i^2 | x_i] = \sigma_i^2$.
- ▶ In the **homoskedastic** case $\mathbf{D} = \sigma^2 \mathbf{I}_n$.

$$\mathbf{D} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2) = \mathbb{E}(\varepsilon_i \varepsilon_i' \mid \mathbf{X}) = \mathbb{E}(\tilde{\mathbf{D}} \mid \mathbf{X})$$

We can estimate $\hat{\mathbf{V}}_{\hat{\beta}}$ by plugging in $\tilde{\mathbf{D}} \rightarrow \mathbf{D}$:

$$\mathbf{v}_{\hat{\beta}} = (X'X)^{-1}(X'\tilde{\mathbf{D}}X)(X'X)^{-1} = (X'X)^{-1} \left(\sum_{i=1}^N x_i x_i' \varepsilon_i^2 \right) (X'X)^{-1}$$

The expectation shows us this estimator is unbiased:

$$\begin{aligned} \mathbb{E}[\mathbf{v}_{\hat{\beta}} \mid X] &= (X'X)^{-1} \left(\sum_{i=1}^N x_i x_i' \mathbb{E}[\varepsilon_i^2 \mid X] \right) (X'X)^{-1} \\ &= (X'X)^{-1} \left(\sum_{i=1}^N x_i x_i' \sigma_i^2 \right) (X'X)^{-1} = (X'X)^{-1}(X'\mathbf{D}X)(X'X)^{-1} \end{aligned}$$

Heteroskedasticity Consistent (HC) Variance Estimates

What we need is a consistent estimator for $\hat{\varepsilon}_i^2$.

$$\mathbf{v}_{\hat{\beta}}^{HC0} = (X'X)^{-1} \left(\sum_{i=1}^N x_i x_i' \hat{\varepsilon}_i^2 \right) (X'X)^{-1}$$

$$\mathbf{v}_{\hat{\beta}}^{HC1} = (X'X)^{-1} \left(\sum_{i=1}^N x_i x_i' \hat{\varepsilon}_i^2 \right) (X'X)^{-1} \cdot \left(\frac{n}{n-k} \right)$$

Could use leave one out variance estimate:

$$\mathbf{v}_{\hat{\beta}}^{HC2} = (X'X)^{-1} \left(\sum_{i=1}^N (1 - h_{ii})^{-1} x_i x_i' \hat{\varepsilon}_i^2 \right) (X'X)^{-1}$$

$$\mathbf{v}_{\hat{\beta}}^{HC3} = (X'X)^{-1} \left(\sum_{i=1}^N (1 - h_{ii})^{-2} x_i x_i' \hat{\varepsilon}_i^2 \right) (X'X)^{-1}$$

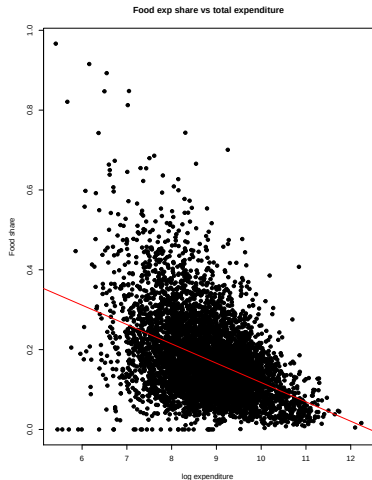
Heteroskedasticity Consistent (HC) Variance Estimates

- ▶ We know that $\mathbf{v}_{\hat{\beta}}^{HC3} > \mathbf{v}_{\hat{\beta}}^{HC2} > \mathbf{v}_{\hat{\beta}}^{HC0}$ because $(1 - h_{ii}) < 1$.
- ▶ *HC3* are the most **conservative** and also place the most weight on potential outliers.
- ▶ Stata uses *HC1* as the default and it is what most people refer to when they say **robust standard errors**.
- ▶ These are often called White (1980) SE's or Eicher-Huber-White SE's.
- ▶ In small sample some evidence that *HC2* has better **coverage**, (what is that?)

Example: Engel Curves

- ▶ Engel curves refer to the relationship between a household's expenditure share on a good and income (or total expenditure).
- ▶ Engel curves for food are typically downward sloping – as total expenditure of a household increases, the proportion of its expenditure dedicated to food falls.
 - Expenditure on food still rises as total expenditure rises, but less than proportionally, so that food's expenditure *share* falls.

Food Engel Curves



- If we plotted total food expenditure (rather than the expenditure share), the heteroscedasticity would go in the other direction.

Data source: BLS Consumer Expenditure Survey data

- ▶ Recall that we defined the homoscedasticity assumption as:

$$Var(\varepsilon) = \sigma^2 \mathbf{I}$$

this assumption has two aspects:

1. The disturbance for each observation has the same variance
 2. Imposing zero correlation between disturbances for different observations
- ▶ The terminology can be misleading here, because what people refer to as “heteroscedasticity-robust” standard errors (the variance estimators on the previous slide) are robust to violations of 1 but not 2.
 - ▶ We need to do a bit more to estimate standard errors in a way that is robust to correlated data.

- ▶ The baseline assumptions of the linear regression framework imply that the disturbances are uncorrelated across observations. There are many ways for this to be violated.
 - Example 1: we might have county-level data for a regression and be concerned that different counties within a given state have correlated disturbances because all counties are subject to the same (unobserved) state-level policies.
 - Example 2: time series data (asset prices), and we are worried that some unobserved factors within the disturbances are serially correlated
 - Example 3: county level data again, and we are worried about geographically correlated factors such as weather.

Different correlation patterns call for different estimators of Σ , the variance of $\mathbf{b}_{OLS}$. Some common alternatives to the no-correlation baseline:

1. Clustered standard errors, when there is correlation between observations within well-defined groups, but no correlation between observations in different groups.
2. Newey-West standard errors (and extensions) to deal with serial correlation in time series data.
3. Conley-Newey-West standard errors that allow for correlation in multiple dimensions (especially popular in the context of spatially explicit models).

What is Clustering?

Suppose we want to relax our i.i.d. assumption:

- ▶ Each observation i is a villager and each group g is a village
- ▶ Each observation i is a student and each group g is a class.
- ▶ Each observation t is a year and each entity i is a state.
- ▶ Each observation t is a week and each entity i is a shopper.

We might expect that $\text{Cov}(u_{g1}, u_{g2}, \dots, u_{gN}) \neq 0 \rightarrow$ independence is a bad assumption.

The groups (villages, classrooms, states) are independent of one another, but within each group we can allow for arbitrary correlation.

- ▶ If correlation is within an individual over time we call it **serial correlation** or **autocorrelation**
- ▶ Just like in time-series $\rightarrow$ we have fewer effective independent observations in our sample.
- ▶ Asymptotics now about the number of groups $G \rightarrow \infty$ not observations $N \rightarrow \infty$

- ▶ Suppose data are organized into distinct groups $g = 1, 2, \dots, G$. Let $g(i)$ be the group identity of observation i .
 - e.g., with county-level data, we have $g(\text{Manhattan}) = \text{NY}$.
- ▶ We assume $\mathbb{E} [\varepsilon_i \varepsilon_j] = 0$ as long as $g(i) \neq g(j)$, and we do not restrict the correlation $\mathbb{E} [\varepsilon_i \varepsilon_j]$ for observations within the same group.
- ▶ Intuition: the linear regression framework with no correlation in observations will overstate the precision of our estimates. If we add another observation within a cluster, and that observation is highly correlated with the other observations, it's not actually as good as adding another independent observation.

- Recall the sandwich formula for standard errors:

$$n^{-1} \mathbb{E} [\mathbf{x}_i \mathbf{x}_i']^{-1} \mathbb{V} [\mathbf{x}_i \varepsilon_i] \mathbb{E} [\mathbf{x}_i \mathbf{x}_i']^{-1} .$$

$$\mathbb{V} [\mathbf{x}_i \varepsilon_i] = n^{-1} \sum_i \mathbf{x}_i \mathbf{x}_i' e_i^2$$

- ▶ The cluster-robust estimate of standard errors will be consistent as the number of groups gets large.
- ▶ Note that this estimator adds extra terms (covariance terms) to the estimate of variance, so this is going to make standard errors larger as long as covariances $\mathbb{E} [\varepsilon_i \varepsilon_j]$ are positive.
- ▶ Thus, if standard formulas are used in the presence of cluster-correlated disturbances, standard errors will be too small.
- ▶ Statistical software packages typically make it easy to compute cluster-robust errors.
- ▶ Clustering often makes a **huge** difference in standard errors.

Begin by stacking up observations in each group $\mathbf{y}_g = [y_{g1}, \dots, y_{gn_g}]$, we can write OLS three ways:

$$y_{ig} = x'_{ig}\beta + \varepsilon_{ig}$$

$$\mathbf{y}_g = \mathbf{X}_g\beta + \varepsilon_g$$

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon$$

All of these are equivalent:

$$\hat{\beta} = \left(\sum_{g=1}^G \sum_{i=1}^{n_g} x'_{ig} x_{ig} \right)^{-1} \left(\sum_{g=1}^G \sum_{i=1}^{n_g} x'_{ig} y_{ig} \right)$$

$$\hat{\beta} = \left(\sum_{g=1}^G \mathbf{x}'_g \mathbf{x}_g \right)^{-1} \left(\sum_{g=1}^G \mathbf{x}'_g \mathbf{y}_g \right)$$

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}'\mathbf{Y})$$

Clustering Derivation (Continued)

The error terms have covariance within each cluster g as:

$$\Sigma_g = \mathbb{E} (\varepsilon_g \varepsilon_g' \mid X_g)$$

In order to calculate $\widehat{V}_{\widehat{\beta}}$ we replace the covariance matrix $\mathbf{D}$ with Ω and consider an estimator $\widehat{\Omega}_n$. We exploit **independence across clusters**:

$$\text{var} \left(\left(\sum_{g=1}^G X_g' \varepsilon_g \right) \mid X \right) = \sum_{g=1}^G \text{var} (X_g' \varepsilon_g \mid X_g) = \sum_{g=1}^G X_g' \mathbb{E} (\varepsilon_g \varepsilon_g' \mid X_g) X_g = \sum_{g=1}^G X_g' \Sigma_g \mathbf{x}_g \equiv \Omega_N$$

And an estimate of the variance:

$$V_{\widehat{\beta}} = \text{var}(\widehat{\beta} \mid X) = (\mathbf{X}'\mathbf{X})^{-1} \widehat{\Omega}_n (\mathbf{X}'\mathbf{X})^{-1}$$

$$\widehat{V}_{OLS}^{CR1} = (X'X)^{-1} \left(\sum_{g=1}^G X'_g e_g e'_g X_g \right) (X'X)^{-1}$$

$$\widehat{V}_{OLS}^{CR3} = (X'X)^{-1} \left(\sum_{g=1}^G X'_g \tilde{e}_g \tilde{e}'_g X_g \right) (X'X)^{-1}$$

- Can replace $\mathbf{e}_g \rightarrow \tilde{\mathbf{e}}_g$ for leave-one out like *HC3* (these are called *CR3*).

```
feols(y~ x1 + x2, data=df, vcov=~group_id )  
feols(y~ x1 + x2, data=df, vcov=~group_id+time_id)
```

How should I cluster my standard errors?

- ▶ Heck if I know.
- ▶ This is very problem specific
- ▶ It matters a lot → standard errors can get orders of magnitude larger.
- ▶ Do you believe across group independence or not? [this is the only thing that matters]
- ▶ If you include **fixed effects** probably you need at least clustering at that level.

Newey West Standard Errors (HAC)

- ▶ In serially correlated data we need to account for $\text{Cov}(\varepsilon_t, \varepsilon_{t-1}, \dots) \neq 0$.
- ▶ Clustering is one solution, but we may end up throwing away all of our data.
- ▶ Instead we could estimate the serial correlation.
- ▶ May also want standard errors that are **heteroskedasticity AND autocorrelation consistent** (HAC).
- ▶ Have to select a number of lags L

$$\widehat{\Omega}_{n,L}^{HAC} = \sum_{t=1}^T e_t^2 x_t x_t' + \sum_{l=1}^L \sum_{t=l+1}^T w_l e_t e_{t-l} (x_t x_{t-l}' + x_{t-l} x_t')$$
$$w_l = 1 - \frac{l}{L+1}$$

You will see these constantly in accounting and finance, where serially correlated outcomes are the norm.

What about β ?

- ▶ All of the estimates above should produce **identical** point estimates
- ▶ We have just been talking about adjusting **standard errors**
- ▶ Should the presence of heteroskedasticity change our estimates of $\hat{\beta}$ as well?

- ▶ In principle, if you *knew* the error variances you could reweight observations ($w_i \propto 1/\sigma_i^2$) and beat OLS on efficiency:

$$\hat{\beta}_{WLS} = (X'WX)^{-1}X'Wy \quad (\text{GLS: let } W \text{ be non-diagonal})$$

- ▶ In practice σ_i^2 is unknown; estimating it (FGLS/IRLS) imports a model of the variance into your estimate of β , and misspecifying that model can do real damage.
- ▶ The modern division of labor: **OLS for the point estimate, robust/clustered/HAC standard errors for inference**. That is what every applied paper you read this semester does.
- ▶ The one live exception: **sampling weights** from survey data (your sample is 75% women but the population isn't) — that's a weighting story about representativeness, not about efficiency.

- ▶ Another approach to standard errors entirely: **simulate the sampling distribution** instead of deriving it.
- ▶ Resample your data with replacement, re-estimate, repeat 10,000 times, and take the standard deviation of the estimates.
- ▶ Works for quantities with no textbook formula at all (ratios of coefficients, model predictions, whole counterfactuals).
- ▶ Full treatment — bias correction, confidence intervals, when it fails, variants for clustered data — in part (b) of today's lecture.